

Adaptive Test-Time Compute for Medical Visual Question Answering: More Accurate than a Large Reasoning Model at a Fraction of its Latency and Energy

Anonymous Authors

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All reported figures are computed from measured model outputs; none are fabricated.

Abstract—Deploying medical vision–language models forces a choice between a small, cheap model and a large reasoning model that is far more accurate but an order of magnitude slower and more energy-hungry (a 32B model in reasoning mode costs ≈ 10.5 s and ≈ 2.0 kJ per query at batch 1, versus ≈ 0.35 s for a 7B). We show that test-time compute should be allocated adaptively, and that doing so lets a 7B model beat every fixed way of using the 32B on accuracy while cutting an order of magnitude off its latency and energy. Three empirical findings motivate the design: (1) chain-of-thought reasoning hurts perception-style medical VQA—on prompt- and resolution-matched arms it is strictly worse than answering directly in 17/20 (family $\times$ benchmark) perception cells across five medical model families, pooled $\Delta = -0.040$ (95% CI $[-0.046, -0.035]$, $n=30,250$ paired samples)—while on reasoning-heavy questions the apparent gain is an answer-format effect: with the answer format matched, an explicit reasoning instruction is CI-significant in 0/9 sub-cells, whereas the format contrast is significant in 3/9, because requesting the answer in `\boxed{\}` is itself a reasoning trigger; (2) the answer format also governs whether test-time sampling helps—best-of- N with a trained verifier beats the large model on free-text answers but not on multiple choice; and (3) the escalation signal is fundamentally bounded—no cheap signal reliably predicts which questions the large model will fix (AU-ROC ≈ 0.6 on multiple choice, confirmed by six independent methods). Our format-aware cascade routes multiple-choice questions through a confidence-margin escalation and free-text questions through adaptive best-of- N selection, and adds a small trained answer verifier. We report accuracy as a macro average over the eight evaluation cells of five medical VQA benchmarks (equal weight per cell; $n=42,224$ items, held-out, 10,000-sample bootstrap), because sample-weighting lets one benchmark carry 79% of the suite. On that basis the accuracy-max operating point is $+0.0720$ (95% CI $[+0.0614, +0.0824]$) over the naive always-32B-reasoning baseline at -89% latency and -87% energy, and $+0.0128$ ($[+0.0056, +0.0200]$) over a single always-32B-direct forward; the compute-lean point is $+0.0626$ ($[+0.0514, +0.0734]$) over the reasoning baseline. We are explicit about the cost accounting, because it does not point the same way as the accuracy: sample-weighted—the cost a deployment on this traffic mix would actually pay—the two operating points use $0.49\times$ and $0.93\times$ the compute of one 32B forward, but under equal weight per cell they use $1.20\times$ and $1.41\times$, because the compute saving is concentrated

on the low-escalation multiple-choice cells rather than being uniform. The defensible joint claim is therefore large latency and energy savings against a reasoning baseline, plus real but non-uniform compute savings.

Index Terms—medical VLM, model cascade, test-time scaling, best-of- N , verifier, inference efficiency, learning to defer

I. Introduction

A medical vision–language model (VLM) maps an image—a radiograph, a pathology slide, an endoscopy frame—together with a natural-language question to an answer. The most accurate open medical VLMs are large “reasoning” models that emit an explicit chain-of-thought inside `<think>` tags before answering. Reasoning improves the hardest questions, but it is expensive: for the Lingshu-32B model we measure a batch-1 latency of 10.5s and 2.0kJ of GPU energy per query in think mode, versus 0.35s for the Lingshu-7B model. In a clinical setting—triage, second-read, point-of-care—this cost gap is the central obstacle to deployment.

The standard remedy is a cascade: answer with a cheap model, and escalate only the hard cases to the expensive one. A cascade’s usefulness hinges on two decisions: which queries to escalate (the gate), and what to run once escalated (the action). Prior cascading work concentrates on the gate. We find that, for medical VQA, the gate is the least improvable part of the system, and that the large gains come from a more careful choice of action, informed by three empirical findings.

Finding 1: reasoning over-thinks perception, and on reasoning-heavy questions the operative lever is the answer format. A `<think>` trace lowers accuracy on perception-style medical VQA (organ identification, presence/absence, modality). This holds across all five medical model families we test, on prompt- and resolution-matched arms, and it survives restriction to arms that differ by nothing but the reasoning instruction. On genuinely reasoning-heavy questions, getting a reasoning-

tuned model to emit a trace does raise accuracy substantially, but a matched-prompt re-run shows the trace is elicited by the answer-format instruction rather than by the reasoning instruction: requesting the answer in `\boxed{\}` makes MedVLThinker-32B and InternVL3-38B generate hundreds of reasoning tokens with no reasoning trigger present at all. We therefore report the reasoning-side result as an answer-format effect and do not claim that a reasoning instruction improves accuracy. Consequently, the large model’s fast no-think mode is an equal-or-better—and $16\times$ cheaper—action than its slow think mode on most of the workload.

Finding 2: the answer format decides whether sampling helps. On free-text answers, drawing N samples from the small model and selecting one with a trained verifier is highly effective (the correct answer is usually among the samples, and the verifier finds it). On multiple-choice questions the same procedure fails: the discriminative signal lives in the option set, and collapses when a model must generate rather than choose.

Finding 3: the escalation signal is bounded. A cascade needs to predict recoverability—will the strong model fix this error?—not merely cheap-model wrongness. On multiple choice this is nearly unpredictable from any training-free signal (AUROC ≈ 0.6), because the two models’ errors are correlated. We confirm this bound with six independent methods; no decision rule materially beats a plain confidence threshold.

These findings yield a format-aware adaptive cascade. Multiple-choice questions take a confidence-margin escalation to the large model’s no-think mode; free-text questions take adaptive best-of- N from the small model, selected by a trained verifier, with a residual escalation. Because reasoning fires only on the small genuine-reasoning residual and free-text answers are handled cheaply, the system is far cheaper than always running the large model, at equal or higher accuracy.

Contributions.

- C1 A systematic study of when test-time compute helps in medical VQA, establishing findings 1–3 with confidence intervals and cross-family validation (§V-A).
- C2 A format-aware adaptive cascade that is more accurate than every fixed single-model strategy—including an oracle mode-selection baseline—under an equal-weight-per-benchmark-cell average, at an order of magnitude less latency and energy than the reasoning baseline, together with an honest account of where its compute saving does and does not generalise (§IV, §V-B).
- C3 A small trained outcome verifier that makes small-model best-of- N competitive-with-or-better than the large model on open-ended medical VQA—provisional, pending a disjoint-split retrain (see Limitations)—coupled to an optimal-stopping controller that cuts free-text sampling cost by 33% at equal accuracy (§IV, §V-C).

- C4 A rigorous, multiply-confirmed characterization of the fundamental limits—a recoverability bound on escalation and a selection bound on verification—that explains why “a better gate” is the wrong lever (§V-D).

All numbers are computed from measured model outputs; no figure is fabricated, and every headline comparison is reported with a paired-bootstrap confidence interval.

II. Related Work

LLM/VLM cascades and routing. FrugalGPT [1] escalates through a chain of models by a learned post-generation confidence. Agreement-based cascading (ABC) [2] escalates on ensemble disagreement, and certainty-adaptive routing (CAR) [3] gates reasoning by a certainty estimate; RouteLLM [4] and Hybrid-LLM [5] learn a query router between a weak and a strong model. Our multiple-choice gate is deliberately in this family—we do not claim it novel. Our contribution is the format-aware structure it controls and the choice of action (the large model’s no-think mode as an intermediate, verifier-selected best-of- N for free text).

Confidence, selective prediction, and deferral. Maximum softmax probability [6], [7], entropy, and split-conformal routers [8] are the training-free gate family we benchmark. Learning-to-defer [9]–[11] formalizes when to hand a query to a second predictor; Jitkrittum et al. [10] show confidence-based deferral is optimal only when the strong model’s error is well predicted by cheap-model confidence—exactly the condition that fails here.

Test-time scaling and best-of- N . Self-consistency [12] and best-of- N sampling improve reasoning by drawing multiple candidates; a verifier or reward model then selects among them. Generative verifiers (GenRM) [13] cast reward modeling as next-token prediction, and $P(\text{True})$ self-scoring [19] lets a model estimate its own correctness. Two facts shape our use of these tools. First, the benefit is format-dependent: selection helps free-text answers, where candidates are diverse and a verifier can rank them, but not multiple choice, where a handful of options makes the “candidate set” degenerate (Finding 2). Second, the number of samples should itself be adaptive; we control it with Weitzman’s optimal-search rule [14], a classical result in the economics of sequential search that, to our knowledge, has not previously been used to schedule test-time sampling.

Efficient VLMs and medical VLMs. Visual-token pruning (e.g., FastV [16]) and low-bit quantization reduce per-forward cost orthogonally to routing; because our compute axis is precision-independent MAC-count, quantization is a memory/energy lever that leaves the FLOP frontier unchanged. Medical reasoning VLMs such as Lingshu [17] and MedVLThinker [18] are strong open baselines with explicit think modes, and are our test beds. Prior medical generator-verifier pipelines target training-data synthesis; we instead deploy a trained verifier at inference time.

Our verifier is a constructive counterpoint to reports that self-verification fails in medical VQA: zero-shot self-verification indeed does not help, but a small trained outcome verifier does, and it is exactly the component that breaks the free-text selection floor (§V-D).

Reasoning cost. A separate line questions whether chain-of-thought always helps; we contribute a medical-VQA-specific, cross-family measurement that it hurts perception (Finding 1), which is what makes the fast no-think mode the right escalation action rather than a compromise.

III. Experimental Setup

Models. The cheap leg is Lingshu-7B [17] run no-think and greedy; the strong leg is Lingshu-32B in no-think and think modes. The verifier is a LoRA head on a Lingshu-7B base. Finding 1 is additionally validated on MedVLThinker-7B/32B [18], InternVL3-8B/38B, QoQ-Med-VL, Chiron-o1, and MedGemma, with InternVL2.5-8B and Phi-3.5-Vision as non-medical controls.

Prompt matching for Finding 1. Because a think-vs-no-think comparison is only as good as the pairing behind it, every cell in Finding 1 uses the most closely matched pair of dumps available for that family, and we record for each pair whether the image budget is identical, whether the answer-format constraint is present in both arms, whether the think arm carries a persona the direct arm lacks, and—as a positive control that the manipulation took effect—the mean number of generated tokens in each arm. Two consequences are worth stating because they are easy to get wrong. First, a “reasoning” prompt that does not actually elicit reasoning produces a null comparison that can be mistaken for a null result: one family’s published native-think configuration turned out to be an answer-format instruction only, emitting 3.0 generated tokens, and we replace it with an arm that emits 150–259. Second, matching the arms strengthens the perception effect rather than weakening it (from 15/20 to 17/20 strictly negative cells), so the reported result is not an artifact of the correction.

Benchmarks. We evaluate with MedEvalKit on five medical VQA benchmarks (Table I), split two ways. By regime: perception (PMC-VQA, SLAKE, VQA-RAD, PathVQA) versus reasoning (MedXpertQA-MM). By format: multiple-choice (MCQ, $n=39,879$) versus free-text open-ended (SLAKE/VQA-RAD/PathVQA open splits, $n=2,345$); three benchmarks carry both, giving eight evaluation cells. The pooled suite is $n=42,224$. Open-ended answers are graded by exact/normalized match with a neutral-LLM adjudication of near-misses.

PMC-VQA split (reproducibility). PMC-VQA ships more than one test file, and they are not interchangeable, so we state ours explicitly: all PMC-VQA results in this paper are on test_2.csv, the version-2 (“noncompound images”) test split of 33,430 items, which is the file MedEvalKit loads and therefore the split the published Lingshu baselines we compare against were scored on.

This is not the authors’ 2,000-item human-verified split (test_clean.csv, version 1); the two files intersect on only 6 items, so results on one cannot be mapped onto the other. We use test_2.csv for comparability with prior work and note that no verification pass or reference accuracy has been published for it.

Faithful reproduction. We reproduce the published Lingshu accuracies within one point on every benchmark in the suite (7B: SLAKE 82.5 vs. 83.1; PMC-VQA 54.3 vs. 56.3, both on test_2.csv; MedXpertQA 26.2 vs. 26.7), confirming the evaluation harness is faithful before any cascade claim is made.

Cost model. A single forward pass costs $F = 2 \Theta (P + G)$ multiply-accumulates (MACs), where Θ is the parameter count, P the prompt tokens (including vision tokens), and G the generated tokens. We report compute in FLOP-equivalents normalized to one 7B forward; the measured 32B/7B ratio is 4.57. Because MACs are precision-independent, low-bit quantization does not change this axis. For a cascade with per-tier cost c_t and escalation probabilities e_0 (past tier 0) and e_1 (to the reasoning tier),

$$C = c_0 + e_0 c_1 + e_1 c_2. \quad (1)$$

Latency and energy use (1) with per-tier costs from real batch-1 measurements (NVML-integrated energy): a 7B no-think pass is 347 ms/45.8 J (1.0 FLOP-eq); a 32B no-think pass 665 ms/127 J (4.57); a 32B think pass 10,521 ms/2,002 J (4.57, a decode-FLOP lower bound); a verifier pass scoring all N candidates 175 ms (1.0). All accuracy comparisons are paired and use a 10,000-resample bootstrap over questions for 95% confidence intervals.

Verifier training. The outcome verifier is a low-rank (LoRA) head on a frozen Lingshu-7B, trained to predict a binary correctness label from (image, question, candidate answer). Training data are held-out medical VQA answers pooled from four open-ended families (SLAKE, VQA-RAD, PathVQA, Kvasir), with correctness labels from the same grader used at evaluation. At inference one batched forward scores all N candidates. All verifier accuracies are reported on a disjoint held-out split ($n=1,064$); its discrimination AUROC is measured on a larger pool ($n=8,512$).

Reporting convention: equal weight per benchmark cell. All headline accuracies are macro averages over the eight evaluation cells, each weighted 1/8. We do not headline a sample-weighted (pooled) average, because the cell sizes are extremely unequal: PMC-VQA alone is 33,430 of the 42,224 items, so pooling gives one benchmark 79.2% of the suite and effectively lets it speak for the method. Equal weighting moves PMC-VQA to 12.5% of the weight, the three free-text cells from 5.6% of the items to 37.5% of the weight, and the five multiple-choice cells from 94.4% to 62.5%. Two consequences must be stated up front. First, the confidence intervals get wider (2–4 $\times$), because small benchmarks now count as much as the large one; we regard

that as the honest price of the convention and keep every small cell in, with its own interval. Second, the macro interval covers question-sampling noise within cells but not dataset-selection noise—with eight units, resampling datasets is hopelessly unstable—so we accompany every headline comparison with a leave-one-cell-out range, the macro delta recomputed with each cell dropped in turn. As a robustness check we also compute a five-benchmark macro (a benchmark’s closed and open cells averaged into one number first, 1/5 each), which is the strongest counter-convention a reader can ask for, since three benchmarks contribute a closed and an open cell drawn from the same images.

Cost is reported under both weightings, and they disagree. Compute, latency and energy are additive per query, so a sample-weighted cost is what a deployment serving this traffic mix would actually pay, whereas an equal-weight cost asks the different question of whether the saving generalises across task types. Since the two answers differ in sign here, we report both and label each; we never pair a macro accuracy with a sample-weighted cost.

Statistics. Every reported delta is a paired difference (same questions, both systems) with a 10,000-resample bootstrap over questions for the 95% CI; “significant” means the CI excludes zero. Thresholds (τ , the reservation value, the escalation and veto cutoffs) are frozen by nested cross-fitting on training folds, so the intervals reflect question-sampling noise with the policy held fixed. One mis-specification we state rather than hide: those thresholds were calibrated against a pooled iso-accuracy target, while the report is now macro; a macro-objective refit is a separate experiment we have not run, so the numbers below are a pooled-tuned policy scored on an equal-weight metric.

Baselines. We compare against (i) always-32B-think, the naive “use the big reasoning model for everything”; (ii) always-32B-no-think, a strong single-forward reference; (iii) oracle mode-selection, which picks, per benchmark, whichever of think/no-think is more accurate and pays that mode’s cost—a non-deployable upper bound on any single-32B strategy; and (iv) always-7B, the cheap floor.

IV. Method

The system is a two-arm router (Fig. 1). A format detector reads the prompt (never the gold answer): questions with an explicit option set take the MCQ arm; free-text questions take the open-text arm. Both arms escalate to the same strong leg—the 32B in no-think mode. This is a direct consequence of Finding 1 (§V-A): reasoning helps neither perception MCQ nor free-text perception, so the slow think pass is invoked only for genuine reasoning, which on this workload is a negligible residual. The always-32B-think baseline is therefore paying a 16× cost for a mode that rarely helps.

TABLE I

The evaluation suite. Perception vs. reasoning regime; formats present; sample counts. Total: 39,879 MCQ + 2,345 open = 42,224. [†]PMC-VQA is the version-2 test split test_2.csv (33,430 items), the file MedEvalKit loads—not the 2,000-item human-verified test_clean.csv (see §III).

Benchmark	Regime	#MCQ	#Open
PMC-VQA [†]	perception	33,430	—
PathVQA	perception	3,362	1,500
SLAKE	perception	836	645
VQA-RAD	perception	251	200
MedXpertQA-MM	reasoning	2,000	—
Total		39,879	2,345

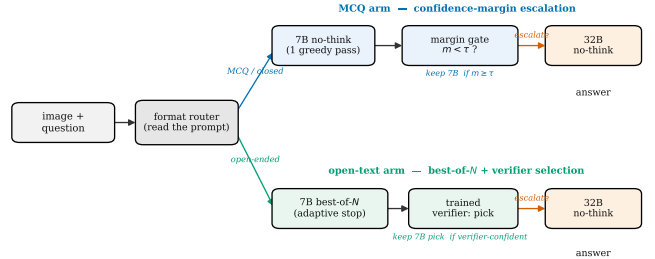


Fig. 1. The format-aware adaptive cascade. A prompt-based format detector routes to a confidence-margin escalation (MCQ) or to adaptive best-of- N with a trained verifier (open-text). Both arms escalate a low-confidence residual to the 32B in fast no-think mode. The method always answers—the veto and deferral rules choose which model answers, never abstain.

A. MCQ arm: confidence-margin escalation

The cheap model produces per-option log-probabilities $\ell_1 \geq \ell_2 \geq \dots$ with softmax $p_i = e^{\ell_i} / \sum_j e^{\ell_j}$. The escalation signal is the margin

$$m = p_1 - p_2, \quad (2)$$

the gap between the top two option probabilities: a large margin means the cheap model is confidently committed to one option. We escalate (re-answer with the 32B) iff $m < \tau$. The threshold τ is frozen by 5-fold cross-fitting: on each training fold we take the minimum-escalation τ whose cascade accuracy reaches the strong leg’s accuracy (an iso-accuracy target), evaluate on the held-out fold, and average. This targets parity at the least escalation, so the cascade never underperforms the strong leg while escalating a minority of queries.

B. Open-text arm: adaptive best-of- N with a trained verifier

Free-text answers have no option logprobs, so the margin gate has no analog. Instead the cheap model draws N candidate answers; a trained outcome verifier $V(\text{image}, \text{question}, \text{candidate}) \in [0, 1]$ scores each (all N in one batched forward), and we select $\arg \max_i V(\cdot)$. If the top verifier score is low we escalate to the 32B. The verifier is a LoRA head trained on held-out medical VQA

Algorithm 1 Format-aware adaptive cascade (one query)

```

1: input: image  $x$ , question  $q$ ; thresholds  $\tau, z, v_{\text{esc}}$ 
2: if  $q$  is multiple-choice then
3:    $(\hat{a}_7, p_1, p_2) \leftarrow \text{Cheap}_{\text{nt}}(x, q)$ ;  $m \leftarrow p_1 - p_2$ 
4:   if  $m \geq \tau$  then
5:     return  $\hat{a}_7$  {keep cheap answer}
6:   else
7:     return  $\text{Strong}_{\text{nt}}(x, q)$  {escalate}
8:   end if
9: else
10:  candidates  $\leftarrow \emptyset$ ;  $v^* \leftarrow 0$ 
11:  while  $v^* < z$  and  $|\text{candidates}| < N_{\text{max}}$  do
12:    draw  $a \sim \text{Cheap}_{\text{nt}}(x, q)$ ; add to candidates
13:     $v^* \leftarrow \max_i V(x, q, a_i)$ 
14:  end while
15:   $\hat{a}_7 \leftarrow \arg \max_i V(x, q, a_i)$ 
16:  if  $v^* \geq v_{\text{esc}}$  then
17:    return  $\hat{a}_7$ 
18:  else
19:    return  $\text{Strong}_{\text{nt}}(x, q)$ 
20:  end if
21: end if

```

to predict answer correctness; it discriminates correct from incorrect candidates at AUROC 0.924 ($n=8,512$; §V-C).

How many samples? An optimal-stopping controller. Drawing a fixed $N=8$ wastes compute on easy questions and under-samples hard ones. We choose N per question with Weitzman’s optimal-search rule [14]. Frame each additional draw as opening a box: opening costs λ (one 7B forward, in FLOP-equivalents, weighted by the value of a correct answer) and returns a reward equal to the verifier score V of the new candidate, drawn from a distribution F . Weitzman’s reservation value z is the score that makes a marginal draw break even,

$$\lambda = \mathbb{E}[(V - z)^+] = \int_z^\infty (v - z) dF(v), \quad (3)$$

and the optimal policy is: keep drawing while the best score so far v^* is below z , and stop as soon as $v^* \geq z$ (equivalently, stop when the expected marginal improvement $\mathbb{E}[(V - v^*)^+]$ falls below the draw cost λ). The single free parameter λ is tuned on the training folds to hold the fixed- $N=8$ accuracy at the fewest draws. This cuts the mean draw count from 8 to 4.3—a 33% reduction in free-text sampling FLOPs—at held-out iso-accuracy (§V-C). (Draws within a question are correlated, mildly violating Weitzman’s independence assumption; the tuning of λ absorbs this in practice.)

C. Two operating points on one knob

The method exposes a single accuracy/compute knob with two reference settings.

Compute-lean runs Algorithm 1 as written and is tuned for minimum compute at parity.

Accuracy-max adds two certified accuracy levers where a held-out guardrail proves they help.

(i) Confidence-advantage fusion. On the largest MCQ slice (PMC-VQA) the cheap and strong models are comparably skilled with de-correlated errors, so fusing their calibrated per-sample decisions beats either alone. Treating each model $k \in \{7, 32\}$ as a detector that emits a decision u_k with calibrated per-sample reliability $q_k = P(u_k \text{ correct})$, the optimal (Chair–Varshney) decision fusion [15] accepts the answer maximizing the summed log-reliability weight

$$\hat{a} = \arg \max_a \sum_{k: u_k = a} \log \frac{q_k}{1 - q_k}, \quad (4)$$

which, under an equal option count, reduces to trusting whichever model holds the larger calibrated confidence advantage on that item. Because the two models must both be run, this cell costs one extra 7B forward but keeps batch-1 latency at parity with the 32B (the legs run concurrently).

(ii) Certified confidence veto. In the compute-negative variant, fusion is replaced by a one-sided certificate. The cheap model runs on every item; the calibration set is partitioned into cells c by (benchmark $\times$ cheap-confidence-bin), and in each cell we compute the Wilson lower confidence bound $\underline{p}_c$ on cheap-model precision from $\hat{p}_c$ over n_c items,

$$\underline{p}_c = \frac{\hat{p}_c + \frac{z^2}{2n_c} - z \sqrt{\frac{\hat{p}_c(1-\hat{p}_c)}{n_c} + \frac{z^2}{4n_c^2}}}{1 + z^2/n_c}, \quad z=1.96. \quad (5)$$

Where $\hat{p}_c \geq \text{acc}_{32}$ the cheap answer is kept (the 32B is “vetoed”); elsewhere the 32B is taken. Because the bound is one-sided, the result is never-worse than the 32B by construction, and—since vetoed cells skip the 32B—it is cheaper than always-32B.

(iii) Team-optimal open-text deferral. For the open-text arm, a rejector $r(x) \in \{0, 1\}$ over cheap-side features replaces the parity-targeting escalation gate. It is trained to minimize the team risk

$$\min_r \mathbb{E}[(1 - r) \ell(\hat{a}_7) + r \ell(\hat{a}_{32}) + r \kappa], \quad (6)$$

i.e. it escalates only when the expected 32B improvement exceeds the escalation cost κ ; unlike the parity gate, its objective is the team accuracy, which lifts the open-text arm above the 32B on the two slices where the parity gate merely tied.

Concurrency. The 32B image-prefill does not depend on the cheap output, so on the MCQ arm it is launched concurrently with the cheap pass. Splitting the 32B cost into a prefill fraction ϕ and a decode fraction $1-\phi$, an escalated query pays

$$c_{\text{esc}} = \max(c_{\text{cheap}}, \phi c_{32}) + (1-\phi) c_{32} \quad (7)$$

instead of $c_{\text{cheap}} + c_{32}$. The cheap pass hides entirely under the 32B prefill whenever $\phi \geq c_{\text{cheap}}/c_{32}$; with the measured

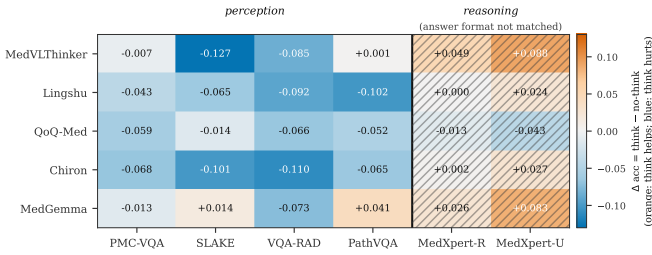


Fig. 2. Finding 1: reasoning over-thinks perception. Accuracy change from adding a `<think>` trace (think – no-think) for five medical model families, on the best prompt- and resolution-matched arms available for each family. Blue = thinking hurts; orange = thinking helps. Perception benchmarks (left of the divider) are blue in 17/20 cells (14/20 with 95% CIs excluding zero). The reasoning block on the right is hatched because it is not a reasoning gain: those arms differ in answer format as well as in the reasoning instruction, and a format-matched re-run attributes the effect to the format (0/9 trigger effects CI-significant vs. 3/9 format effects; §V-A). It is shown for completeness and must be read as “trigger + format”. The third reasoning cell counted in Table II, MMMU-Medical, is omitted here because it is excluded from the evaluation suite on contamination grounds (Limitations (ix)). Source: finding1_corrected_2026-07-29.json, policy P1_audit_best_matched.

$\phi=0.586$ (and $c_{\text{cheap}}/c_{32} = 347/665 = 0.52$) this holds, reducing batch-1 escalation latency by $\approx 12\%$ at identical accuracy.

Every rule answers. The margin gate, the certified veto, and the deferral rule all choose which model produces the answer; none abstains or defers to a human. The method always returns an answer.

Reproduction. The full pipeline is one command (python3 src/cascade_methods/method_final.py), which recomputes every headline number from the per-sample model outputs without a GPU.

V. Results

A. The three findings

Finding 1: chain-of-thought hurts perception; on reasoning-heavy questions it helps only some models. On multiple-choice perception VQA the effect is cross-family and it is measured on matched arms (Fig. 2, Table II): across five medical families, 17 of 20 perception cells have think < no-think strictly, 14 of 20 with 95% paired-bootstrap CIs excluding zero, and 19 of 20 no better than +0.02; the pooled effect is $\Delta = -0.0401$ [-0.0456, -0.0347] over 30,250 paired samples. VQA-RAD is negative in all five families. The count is insensitive to how the arms are matched: three independent matching policies (best-matched arm; strict resolution and answer-format matching; strict with MedVLThinker matched at full rather than reduced resolution) all give 17/20 and a pooled Δ between -0.0401 and -0.0405 . Restricting to the subset of cells whose two arms differ by nothing but the reasoning instruction reproduces it at the same strength: 6/8 strictly negative for Chiron-o1-8B and MedGemma-27B, pooled -0.0273 [-0.0367, -0.0176] ($n=12,100$), and 7/8 for the two

non-medical control architectures. Because multiple-choice gold is a single letter graded by exact equality, the only channel an unmatched prompt has is answer extraction failure; we measure that at 0–3.4% per cell, and charging every extraction failure against the finding changes no count. The one genuine exception is MedGemma-27B on PathVQA, +0.0413 [+0.0220, +0.0607], which survives full prompt matching and which we therefore report as a real exception rather than an artifact.

On reasoning-heavy benchmarks the effect is the answer format, not the reasoning instruction. Taken at face value, 12 of 15 internal-harness cells are point-positive with 4 CI-significant, carried by MedVLThinker-32B (3/3 significant: MMMU +0.0882, MedXpertQA-R +0.0491, MedXpertQA-U +0.0884) and MedGemma-27B (3/3 positive, 1/3 significant), and corroborated on a second harness by MedVLThinker-32B (MMMU +0.100 [+0.027, +0.173]) and InternVL3-38B (MMMU +0.120 [+0.047, +0.193]). But every one of those contrasts pairs a “direct” arm that requests a bare answer letter against a “reasoning” arm that requests both a reasoning trace and the answer in `\boxed{\}`. We therefore ran the missing third arm—`\boxed{\}` requested, no reasoning trigger—on the second harness for the three families with reasoning dumps, holding the seed, decoding parameters, image budget and token limit fixed, and pairing on item id (9 sub-cells; $n=145$ for MMMU multiple-choice-only, 1,446 and 554 for the two MedXpertQA splits). This decomposes each published delta into an answer-format component and the marginal contribution of the reasoning instruction. The result is unambiguous: 0 of 9 reasoning-trigger effects have a 95% CI excluding zero (8/9 are point-positive; the mean shift from matching is -0.028), whereas 3 of 9 answer-format effects do (MedVLThinker MedXpertQA-R +0.046 [+0.019, +0.072] and MedXpertQA-U +0.043; InternVL3 MMMU +0.090). Per cell, published / format / trigger: MedVLThinker MMMU +0.103 / +0.062 / +0.041; InternVL3 MMMU +0.124 / +0.090 / +0.035; Lingshu-32B MMMU +0.028 / -0.014 / +0.041, and ≈ 0 on both MedXpertQA splits for all three components. Answer extraction succeeded on at least 99.86% of items in every new arm (exactly 100% in six of the nine), so none of this is a parsing artifact.

The mechanism is that requesting `\boxed{\}` is itself a reasoning trigger. With no reasoning instruction present, MedVLThinker-32B generates 431–580 tokens on 99–100% of items and InternVL3-38B 193–289 tokens on 94–95%; Lingshu-32B never does (3–4 tokens). We therefore do not claim that a reasoning instruction improves accuracy on reasoning-heavy medical VQA. The supported claim is the weaker one that getting a reasoning-tuned model to emit a trace raises accuracy substantially, with the answer format as the operative lever. Because no prompt in these families suppresses the trace without also changing the answer format, a clean reasoning-vs-no-reasoning contrast

is not obtainable here; we instead report the monotone ladder, which is what a gated reasoning tier actually buys: MedVLThinker-32B on MMMU (multiple-choice only) scores 0.634 at 2 generated tokens, 0.697 at 431, and 0.738 at 580. Lingshu-32B—the family the cascade is built on—is not evidence that reasoning helps under any pairing: with a genuinely reasoning arm against a genuinely direct one it gains +0.041 on MMMU (not significant) and ≈ 0 on both MedXpertQA splits. QoQ-Med-VL-32B is likewise not reasoning-side evidence (MedXpertQA-U significantly negative, -0.0433 , $p=0.022$). None of this changes the cascade’s design or the value of its gated reasoning tier—the rung-1-to-rung-3 total is what such a tier delivers—only the attribution. It does change the methodology we recommend: any think-vs-direct comparison must be answer-format matched and generated-token audited, since a “direct” arm emitting hundreds of tokens is not a direct arm. Crucially, the harm on perception comes with a large cost: a think pass is 15–49 $\times$ slower than a no-think pass for models with a real think mode (Table II, last column). Implication: the large model’s fast no-think mode is the right escalation target for almost all of the workload, and slow reasoning should fire only on the small genuine-reasoning residual—which, for the deployed family, is essentially empty.

Provisional: the free-text version of Finding 1. Adding a <think> trace also appears to lower Lingshu-32B accuracy on free-text perception VQA on all three open sets (SLAKE 0.679 vs. 0.819; VQA-RAD 0.545 vs. 0.600; PathVQA 0.109 vs. 0.376; pooled $\Delta=-0.154$) while costing 16 $\times$ more latency and energy. We mark these magnitudes provisional. The two arms were run with different system prompts—the direct arm additionally carried an expert-analyst persona and the instructions “answer with a short, specific phrase” and “do not explain”, which the reasoning arm dropped. On free-text answers graded against short gold phrases, that is a live answer-style channel rather than the bounded extraction channel of the multiple-choice cells, so part of the gap may be granularity rather than reasoning. A matched-prompt re-run is in progress; the cost side (16 $\times$) is a direct wall-clock and NVML measurement and is unaffected, as is the multiple-choice evidence above. Open-ended figures use the LLM judge (judge_ok), consistent with the main tables; closed and MCQ cells throughout are scored by accuracy.

Finding 2: the answer format decides whether sampling helps. On open-ended answers the test-time signal is strong: a trained verifier discriminates correct candidates at AUROC 0.924, and even a zero-shot confidence signal reaches AUROC 0.85–0.87. On multiple choice the same signals collapse to AUROC ≈ 0.6 . The mechanism is the option set: forcing a single generative verifier to answer PMC-VQA in free-text form (hiding the options) craters accuracy from 0.534 (letter mode) to 0.132 (content mode, $\Delta=-0.394$), and the MCQ correctness signal is un-rankable (AUROC 0.540 on letters). Best-of- N selection therefore

TABLE II

Finding 1, summarized, on prompt- and resolution-matched arms. Δ percep. is the sample-weighted pooled accuracy change from thinking (think – no-think) over the four perception benchmarks, with a 10^4 -resample paired-bootstrap 95% CI ($n=6,050$ per family). “reason. sig.” counts the three reasoning cells (MMMU, MedXpertQA-R, MedXpertQA-U) whose 95% CI excludes zero on the unmatched-format pairing; a format-matched re-run reduces the reasoning-instruction effect to 0/9 CI-significant sub-cells and attributes the gain to the \boxed{} answer format instead (§V-A), so this column should be read as “trigger + format”, not as “reasoning helps”. Thinking hurts perception in four of five families and costs 15–49 $\times$ more where a real think mode exists.

Family	Δ percep.	reason. sig.	think:nt lat.
MedVLThinker	−0.0144	3/3	49 $\times$
Lingshu	−0.0792	0/3	— ^a
QoQ-Med-VL	−0.0524	0/3 ^b	43 $\times$
Chiron-o1	−0.0707	0/3	15 $\times$
MedGemma	+0.0162	1/3	45 $\times$
pooled	−0.0401	4/15	—

95% CIs on Δ percep.: MedVLThinker [−0.0261, −0.0030]; Lingshu [−0.0902, −0.0681]; QoQ-Med-VL [−0.0640, −0.0407]; Chiron-o1 [−0.0840, −0.0577]; MedGemma [+0.0028, +0.0298]; pooled [−0.0456, −0.0347] over all 30,250 paired samples. All cells are multiple-choice, scored by exact letter equality; the free-text version of the finding is reported separately and marked provisional in the text. MedGemma’s positive pooled value is driven by its PathVQA cell (+0.0413 [+0.0220, +0.0607]), the one genuine exception. Arm matching: Chiron-o1 and MedGemma are fully matched (same absent system message, same image budget, answer-format constraint in both arms); MedVLThinker is resolution-matched; Lingshu and QoQ-Med-VL retain a resolution mismatch that no dump on disk can remove—matching it as far as possible strengthens both (−0.0866 and −0.0484 pooled). ^a No matched reasoning; direct latency ratio exists for Lingshu: its previously reported 1.2 $\times$ was measured on an arm whose prompt contained no reasoning trigger (mean 3.0 generated tokens), so it is the ratio of two answer-format prompts rather than of reasoning. ^b QoQ-Med-VL has one significantly negative reasoning cell (MedXpertQA-U, -0.0433 , $p=0.022$).

helps free text and does nothing for MCQ—the two arms of the router are a direct consequence.

Finding 3: the escalation signal is fundamentally bounded. A cascade must predict recoverability: given a cheap-model error, will the strong model fix it? On multiple choice this “recoverability AUROC” sits at ≈ 0.6 for every signal we tried, because the two models’ errors are correlated—agreement between the two no-think legs already carries most of the trust signal ($P(\text{cheap correct} \mid \text{agree})=0.69$ vs. $0.33 \mid \text{disagree}$). We confirm the bound six independent ways (§V-D): decision-level confidence-advantage fusion, a certified veto, Bayesian model averaging, contrastive decoding, full-posterior logit fusion, and an automatic slice-discovery search over 106 candidate slices that certifies zero new beatable slices beyond PMC-VQA (below its label-permutation null). No decision rule beats a plain confidence threshold in a way that generalizes.

External validity of the findings. Finding 1’s perception half is the most robust result in this paper: it holds across five medical families plus two additional general architectures (InternVL2.5-8B pooled -0.0076 and Phi-

3.5-V -0.0187 $[-0.0336, -0.0036]$, both already fully prompt-matched, 7/8 cells strictly negative), it is stable across three arm-matching policies, and it is strongest on the subset of arms with no prompt residual at all. Its reasoning half is weaker and we state it as such: on a format-matched re-run the explicit reasoning instruction is CI-significant in 0 of 9 sub-cells while the answer-format contrast is significant in 3, so the correct claim is an answer-format claim, not a reasoning-instruction claim. Its free-text half is provisional pending a matched-prompt re-run (see above). Finding 2’s direction (open-text signal $\gg$ MCQ signal) reproduces on two base families—the open-text “is-the-cheap-answer-wrong” AUROC is 0.74–0.78 for MedVLThinker-7B and 0.85–0.87 for Lingshu-7B, both far above the MCQ ≈ 0.6 ceiling—though the peak magnitude is calibration-dependent. Finding 3’s trained-verifier lift is positive on two base families (49% of the oracle gap for Lingshu, 25% pooled for a from-scratch MedVLThinker verifier), with the “beats-the-32B” framing being the seed-dependent part.

B. Main result

Table III gives the headline. Under equal weight per benchmark cell, the accuracy-max operating point reaches 0.6694: $+0.0720$ over always-32B-reasoning (95% CI $[+0.0614, +0.0824]$), $+0.0128$ over a single always-32B-direct forward ($[+0.0056, +0.0200]$), and $+0.0122$ over the non-deployable oracle mode-selection baseline ($[+0.0049, +0.0195]$)—all significant. The compute-lean point reaches 0.6600: $+0.0626$ ($[+0.0514, +0.0734]$) over the reasoning baseline, and a tie against always-32B-direct ($+0.0033$, $[-0.0054, +0.0121]$). Both are far above the cheap floor ($+0.0629$ and $+0.0723$ over always-7B).

What the cost axis says, and it is not the same thing. Sample-weighted, the two operating points use $0.49\times$ and $0.93\times$ the compute of one 32B forward—the number a deployment on this traffic mix would actually pay. Under equal weight per cell they use $1.196\times$ and $1.410\times$: no operating point is compute-cheaper than a single 32B forward when every benchmark counts equally. The mechanism is that escalation is extremely heterogeneous across cells—8.5% on PMC-VQA, 20.5% on SLAKE-closed, 45.7% on PathVQA-closed, 57.0% on VQA-RAD-closed, 89.6% on MedXpertQA, and 12.5–35.7% on the free-text cells—and PMC-VQA, the lowest-escalation cell, carries 79.2% of the pooled average; equal weighting raises the multiple-choice escalation rate from 16.2% to 44.2%. Separately, the free-text arm costs 7.6–12.6 FLOP-eq per cell (several 7B draws plus a verifier forward plus escalation) against the baseline’s flat 4.57, while holding 37.5% of the equal weight. We therefore do not claim a uniform compute saving. The claim we do make is: an order of magnitude less latency and energy than a reasoning baseline at higher accuracy, plus a compute saving that is real but concentrated on low-escalation multiple-choice traffic. Against always-32B-reasoning, honestly re-costed, accuracy-max is

-89.0% parallel latency, -78.8% sequential latency and -87.3% energy—but $1.131\times$ its FLOP-eq, so even there the win is latency and energy rather than compute.

Where the equal-weight numbers are load-bearing, and where they fail. Because macro intervals do not cover dataset-selection noise, we report the leave-one-cell-out range for each headline delta: accuracy-max vs. always-32B-reasoning $+0.0720$, range $[+0.0318, +0.0830]$; accuracy-max vs. always-32B-direct $+0.0128$, range $[+0.0023, +0.0146]$; compute-lean vs. always-32B-reasoning $+0.0626$, range $[+0.0225, +0.0732]$. In every case the load-bearing cell is PathVQA-open and the cell holding the claim back is SLAKE-closed. The failure is equally specific: on the five multiple-choice cells alone, compute-lean is significantly worse than simply always running the 32B— -0.0070 ($[-0.0126, -0.0017]$) vs. always-32B-direct and -0.0080 ($[-0.0137, -0.0024]$) vs. oracle-mode—whereas the same comparison is a harmless tie when pooled (-0.0015 , $[-0.0056, +0.0026]$). Accuracy-max repairs it, but only to $+0.0019$ ($[+0.0014, +0.0024]$). On the three free-text cells accuracy-max is $+0.0309$ ($[+0.0116, +0.0502]$) over always-32B-direct and $+0.1848$ ($[+0.1583, +0.2110]$) over always-32B-reasoning. So the equal-weight accuracy win is a free-text win; the multiple-choice half is at best a tie and, at the compute-lean setting, a loss.

Pareto status, stated precisely. The method points remain non-dominated, but because they are more accurate than the 32B strategies, not because they are cheaper: at equal weight, “Pareto-optimal” holds and “Pareto-dominates” does not. Dominance in the strict sense (better on accuracy and every cost axis) survives only against always-32B-reasoning, and there only on accuracy, latency and energy—not on FLOP-eq.

The five-benchmark macro robustness check (§III) moves the accuracy deltas modestly and the sign of nothing: compute-lean $+0.0498$ ($[+0.0402, +0.0594]$) and accuracy-max $+0.0590$ ($[+0.0498, +0.0680]$) over always-32B-reasoning, at $1.098\times$ and $1.310\times$ the compute of one 32B forward.

Table IV shows the per-benchmark decomposition, which under the equal-weight convention is the headline’s decomposition: each of the eight cells contributes exactly 1/8 of the macro average. On MCQ perception the cascade tracks the strong leg within a fraction of a point while escalating 8.5% of PMC-VQA, 20.5% of SLAKE-closed, 45.7% of PathVQA-closed and 57.0% of VQA-RAD-closed. That spread is the reason a single suite-level escalation rate is not a meaningful quantity: it is 16.2% pooled and 44.2% at equal weight over the same five cells. The one benchmark that escalates almost everything (89.6%) is MedXpertQA, where the 7B is near chance and honesty demands paying for the 32B; there the method is slightly below it (-0.0060 , $[-0.0120, -0.0005]$) while costing more, which is a genuine failure mode of a confidence gate in the near-chance regime rather than a rounding artifact.

TABLE III

Main result. Accuracy is the macro average over the eight evaluation cells (equal weight per cell; 42,224 items, held-out, 10^4 -bootstrap; see §III), with the sample-weighted value in the adjacent column for reference. Compute is FLOP-eq ($\times$ one 7B forward) and “rel.” is the ratio to one always-32B forward, reported under both weightings because they disagree in sign. Latency is batch-1 (concurrent / sequential) and energy is NVML-integrated, both macro-weighted. Δ_{reason} vs. always-32B-reasoning; Δ_{direct} vs. a single always-32B-direct forward. Every method point is more accurate than every fixed 32B strategy, but no point is compute-cheaper than one 32B forward at equal weight—the compute saving is real sample-weighted and does not generalise across task types (§V-B). Free-text cells are provisional pending a disjoint-verifier retrain.

System	Acc. (macro)	Acc. (sw)	FLOP -eq	rel. FLOP macro / sw	Lat. (ms)	Energy (J)	Δ_{reason} (95% CI)	Δ_{direct} (95% CI)
always-7B (cheap floor)	0.5971	0.5549	1.00	0.22 $\times$ / 0.22 $\times$	347	45.8	—	—
always-32B-reasoning (naive)	0.5974	0.5591	4.57	1.00 $\times$ / 1.00 $\times$	10,522	2,002	—	−0.0593
always-32B-direct	0.6567	0.5729	4.57	1.00 $\times$ / 1.00 $\times$	665	127	—	0.0000
oracle mode-select 32B	0.6573	0.5730	4.57	1.00 $\times$ / 1.00 $\times$	1,897	361	—	+0.0006
Ours: compute-lean	0.6600	0.5741	5.47	1.196 $\times$ / 0.492 $\times$	650	188	+0.0626 [+0.0514, +0.0734]	+0.0033 [−0.0054, +0.0121]
Ours: accuracy-max	0.6694	0.5836	6.44	1.410 $\times$ / 0.932 $\times$	691	206	+0.0720 [+0.0614, +0.0824]	+0.0128 [+0.0056, +0.0200]
accuracy-max ⁺ (fusion)	0.6661	0.5862	6.56	1.435 $\times$ / 1.250 $\times$	665	209	+0.0686 [+0.0582, +0.0790]	+0.0094 [+0.0013, +0.0176]

The always-32B-reasoning row is charged a flat 4.57 FLOP-eq and 10,522ms by our cost convention, regardless of how much it generates. Honestly re-costed from its measured per-cell generation (3.0–3.3 tokens on four of the five multiple-choice cells, 104–320 on the rest), it is 5.70 FLOP-eq / 6,291ms / 1,625J macro-weighted, against which accuracy-max is −89.0% latency, −87.3% energy and 1.131 $\times$ FLOP-eq. Latency for the free-text arm is reported both concurrent and sequential (1,292–1,350ms sequential for the method rows) because the adaptive draws are sequential by construction and their overlap was never measured.

TABLE IV

Per-benchmark policy and accuracy (compute-lean), against the three single-model baselines, with the paired-bootstrap 95% CI of the method’s delta over the always-32B-no-think reference (all cells real per-sample). “policy”: cheap→32B-nt margin cascade (MCQ) or best-of- N +verifier→32B-nt (open). The cascade matches 32B-no-think within noise on the perception MCQ sets, loses a little only on near-chance MedXpertQA, and wins significantly on PathVQA-open. Under the equal-weight convention (§III) these eight rows carry 1/8 of the headline each, so this table is the headline’s decomposition; the small MCQ shortfalls that are individually within noise are what make compute-lean a significant loss on the multiple-choice half of the macro average (§V-B). The three open rows are provisional pending a disjoint-verifier retrain. n as in Table I. [†]PMC-VQA is test_2.csv (§III).

Benchmark	policy	7B	32B-nt	32B-th	ours	Δ vs 32B-nt	95% CI
PMC-VQA [†]	cascade	0.543	0.552	0.549	0.551	−0.0010	[−0.006, +0.004]
SLAKE-cl	cascade	0.825	0.859	0.864	0.852	−0.0072	[−0.024, +0.008]
VQA-RAD-cl	cascade	0.781	0.853	0.841	0.833	−0.0199	[−0.040, +0.000]
PathVQA-cl	cascade	0.841	0.889	0.889	0.888	−0.0009	[−0.004, +0.002]
MedXpert-MM	cascade	0.262	0.307	0.304	0.301	−0.0060	[−0.012, −0.001]
SLAKE-open	bo- N +V	0.736	0.819	0.679	0.809	−0.0093	[−0.036, +0.017]
VQA-RAD-open	bo- N +V	0.465	0.600	0.545	0.595	−0.0050	[−0.060, +0.050]
PathVQA-open	bo- N +V	0.324	0.376	0.109	0.452	+0.0760	[+0.056, +0.096]

On open-ended perception the method far exceeds always-32B-reasoning: on the free-text pool it scores 0.5625 versus 0.3028 pooled—+0.26 (95% CI [+0.238, +0.281]), or +0.1745 ([+0.1485, +0.2003]) equal-weighted—because reasoning is actively harmful there. The gap is driven by PathVQA-open, where a <think> trace collapses 32B accuracy to 0.109 (−0.267 vs. its no-think mode, as reasoning drifts to plausible-but-wrong pathology answers; verified not a truncation artifact), and by SLAKE-open; the leave-one-cell-out analysis in §V-B confirms PathVQA-open is load-bearing for every headline delta. Three caveats apply specifically to the free-text rows and not to the rest of the table: their two arms were prompt-unmatched in the answer-style direction (see the provisional note in §V-A); the collapse magnitude tracks how terse each set’s gold answers are, so a matched-prompt re-run may reduce it; and the verifier that drives the free-text arm was trained on roughly 70% of the items it is scored on, so all free-text accuracies here are provisional pending a disjoint-

split retrain that is in progress. The vs. always-32B-direct and vs. oracle-mode comparisons do not depend on the reasoning arm at all. This heterogeneity—cheap where the small model suffices, expensive only where it must be—is precisely what a uniform single-model strategy cannot express; it is also, as §V-B shows, exactly why the compute saving does not survive equal weighting.

C. Ablations

Gate bake-off. Table V compares escalation signals on the MCQ pool. Cascade quality (accuracy at a fixed compute budget) tracks recoverability-AUROC, not detection-AUROC: the highest-detection gates (learned-rich 0.693, entropy 0.686, max-probability 0.677) are worse cascades than the plain margin, because detecting cheap-model errors is not the same as detecting recoverable ones. The confidence-margin is the best deployable MCQ gate (15.6% escalation to parity); agreement and resolution-stability rank below it and—being family-specific—do not

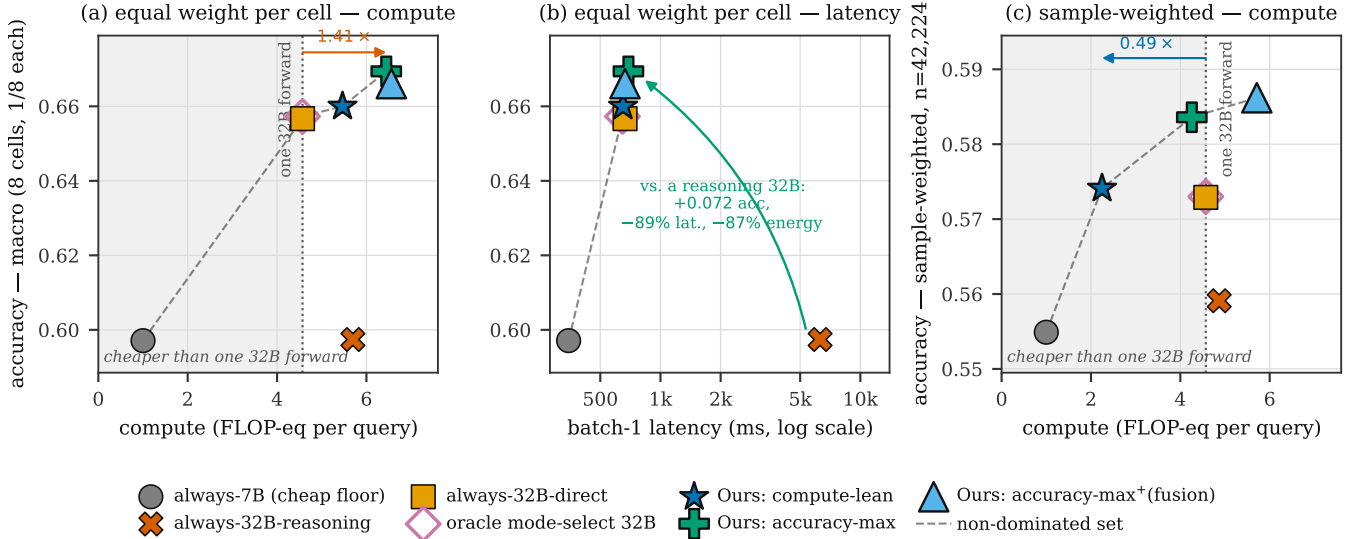


Fig. 3. Accuracy vs. cost, under both weightings. Source: macro_average_headline_2026-07-30.json, key cost.pareto.honest_recost. Each panel pairs an accuracy with a cost computed under the same weighting, since a macro accuracy must never be read against a sample-weighted cost (§III). (a,b) equal weight per reporting cell (8 cells, 1/8 each)—the paper’s primary convention, i.e. the “Acc. (macro)” column of Table III. (c) sample-weighted over the 42,224 items—the “Acc. (sw)” and “rel. FLOP (sw)” columns. The dotted line marks one always-32B forward (4.57 FLOP-eq); the shaded band is the region cheaper than it. The two cost panels disagree, and that is the point. At equal weight no operating point is compute-cheaper than a single 32B forward (1.196 $\times$, 1.410 $\times$, 1.435 $\times$); on this traffic mix two of the three are (0.492 $\times$, 0.932 $\times$). Against a 32B actually made to reason the method wins on every axis but FLOP-eq: +0.0720 accuracy, −89% batch-1 latency, −87% energy, at 1.131 $\times$ its FLOP-eq. The dashed line is the non-dominated set; the method points lie on it because they are more accurate, not because they are cheaper—“Pareto-optimal” holds and “Pareto-dominates” does not (§V-B). Costs are the honest re-cost: always-32B-reasoning and oracle mode-select are charged their measured per-cell generation (5.70 FLOP-eq / 6,291 ms / 1,625 J and 4.57 / 645 ms / 122 J macro-weighted) instead of the flat convention cost carried in Table III’s body; every other point is numerically identical to Table III. Latency is the concurrent batch-1 figure. This figure supersedes fig_pareto_superseded_2026-07-08.pdf, which was sample-weighted only and used an estimated reasoning-baseline accuracy.

TABLE V

Gate bake-off, MCQ pool. Detection AUROC (is the cheap model wrong?) vs. recoverability AUROC (will the 32B fix it?), and escalation to reach 32B-no-think parity. Cascade quality follows recoverability, not detection; the deployed margin gate is best-deployable.

Gate	AUROC _{det}	AUROC _{rec}	esc.%
margin (deployed)	0.725	0.587	15.6
self-verify $P(\text{True})$	0.643	0.614	20.3
max-probability (MSP)	0.677	0.553	20.3
neg-entropy	0.686	0.515	—
learned (grad. boosting)	0.693	0.533	—
agreement (cheap vs. strong)	0.657	—	20.0
resolution-stability	0.724	—	15.5

AUROC_{det}: cheap model wrong? AUROC_{rec}: will 32B fix it? esc.%: escalation to reach 32B-no-think parity.

transfer. On open-text, the trained verifier’s confidence is the best gate (detection AUROC 0.85).

Trained verifier and best-of- N . Table VI evaluates the open-text verifier on $n=1,064$ held-out items. Greedy decoding and self-consistency give 0.41; a same-split single 32B pass gives 0.46; the trained verifier’s best-of- N selection gives 0.50, capturing 49% of the oracle gap (seed 0; 35% on seed 1). The verifier’s selection thus matches or beats the 5 $\times$ -larger 32B (+0.039, 95% CI

TABLE VI

Open-text best-of- N selection ($n=1,064$ held-out). “gap” = fraction of the greedy→oracle gap captured. The trained verifier beats self-consistency and a single same-split 32B pass.

Selector	Accuracy	gap captured
greedy (1 sample)	0.413	0%
self-consistency (majority)	0.411	—
32B single pass (same split)	0.462	—
trained verifier (bo- N)	0.501	35–49%
oracle@ N (upper bound)	0.592	100%

[+0.010, +0.066] on seed 0; a tie on seed 1)—the honest claim is “competitive with the 32B.”

Adaptive sample count. The optimal-stopping controller (§IV) holds this accuracy at fewer draws (Table VII). Across the three open-text benchmarks it cuts the mean draw count from 8 to 4.3 and the free-text sampling cost from 16.2 to 10.9 FLOP-equivalents (−33%) at held-out iso-accuracy, allocating more draws to the harder PathVQA-open (5.5) than to the easier SLAKE-open (3.5). Because the draws are sequential this trades some batch-1 wall-clock for the FLOP and energy saving; the win is compute, not latency.

Accuracy levers. The confidence-advantage fusion im-

TABLE VII

Adaptive best-of- N (optimal stopping) vs. a fixed $N=8$, per open-text benchmark. Mean draws and free-text sampling cost (FLOP-eq) fall by 33% pooled at held-out iso-accuracy.

Benchmark	$\tilde{N}$	$F_{N=8}$	F_{adap}	saved	acc.
SLAKE-open	3.45	16.6	7.6	−54%	0.809
VQA-RAD-open	3.91	16.3	8.4	−48%	0.595
PathVQA-open	5.48	16.0	12.6	−21%	0.452
pooled	4.28	16.2	10.9	−33%	0.563

proves PMC-VQA by +0.0135 over always-32B-no-think (95% CI [+0.010, +0.017], $n=33,430$, test_2.csv)—a genuine broad-slice win on the largest benchmark, where the two models are comparably skilled with de-correlated errors. The certified veto captures 70% of that gain (+0.0095, CI [+0.0071, +0.0118]) while cutting the accuracy-max variant’s compute by 25%—from $1.25\times$ to $0.93\times$ sample-weighted, and from $1.435\times$ to $1.410\times$ at equal weight per cell (vetoed cells skip the 32B). The cut is robust to the weighting; only the sample-weighted sign change is not, and note that at equal weight the veto is also the more accurate of the two levers (+0.0720 vs. +0.0686 over always-32B-reasoning), because PMC-VQA, where the fusion does its work, holds 12.5% rather than 79% of the average. The team-optimal open-text deferral raises PathVQA-open to +0.086 over always-32B-no-think (CI [+0.064, +0.106]) and repairs the two slices where the parity gate tied.

D. The fundamental limits

The recoverability bound. Six independent methods (Table VIII) fail to certify any new MCQ slice beyond PMC-VQA—the one slice where the two models are comparably skilled with de-correlated errors—at which the cheap model can safely replace or improve on the 32B. This includes an automatic slice-discovery search that scans 106 candidate slices per split and certifies zero genuinely-new beatable slices, below its own label-permutation null (1.6 raw certifications vs. 5.6 under shuffled labels). A k -nearest-neighbor recovery gate also fails to beat the scalar margin on all five datasets. The common cause is error correlation: the two models miss the same questions, so “will the 32B fix this?” carries only ≈ 0.6 AUROC of signal regardless of how it is estimated. This is why “a better gate” does not exist as a lever—the ceiling is a property of the model pair, not of the estimator.

The selection bound. Symmetrically, verification is capacity-bounded but the ceiling is not a size artifact. On a pooled open-text selection task, a $7\times$ -larger zero-shot 32B verifier (0.480) ties the small trained 7B verifier (0.475; $\Delta+0.005$, CI [−0.023, +0.032], not significant), whereas the pure-capacity contrast (32B vs. 7B zero-shot) is significant (+0.067). Even the 32B verifier leaves an oracle-to-selection gap of 0.19. Training a small verifier, not enlarging it, is the effective lever—and it is exactly

TABLE VIII

The recoverability bound, confirmed six independent ways. No method certifies a beat-32B MCQ slice beyond PMC-VQA (the one comparable-skill slice); “new cells” counts certified beatable slices beyond PMC.

Method	PMC beat	new cells
confidence-advantage fusion	+0.0135	0
certified confidence veto	+0.0095	0
Bayesian model averaging	+0.0116	0
contrastive decoding	0.000	0
full-posterior logit fusion	n.s.	0
automatic slice discovery (106/split)	—	0

what breaks the selection floor on free text where every training-free selector is stuck.

VI. Discussion and Limitations

We are deliberately explicit about scope. (i) The accuracy edge over the 32B is concentrated: on the three free-text cells (via the verifier) and on PMC-VQA (the largest slice, via fusion or the certified veto). On the small radiology/pathology multiple-choice sets the method matches, not beats, the 32B, and the guardrail keeps it there; under equal weight per cell the compute-lean setting is in fact significantly worse than always running the 32B on the multiple-choice half (−0.0070, [−0.0126, −0.0017]). (ii) The compute saving does not generalise across task types. It is real sample-weighted ($0.49\times$ and $0.93\times$) and absent at equal weight ($1.196\times$ and $1.410\times$), because escalation ranges from 8.5% to 89.6% across cells and the low-escalation cell dominates the pooled average. What survives every weighting is the latency and energy advantage over a reasoning baseline (−89% and −87% honestly re-costed). (iii) The thresholds were cross-fit against a pooled iso-accuracy objective while the headline is now an equal-weight average; a macro-objective refit is future work, so these are pooled-tuned numbers scored on a macro metric. (iv) The free-text accuracies are provisional: the verifier was trained on a grouped 70/30 split of the same four open-ended families and then scored over the full sets, so roughly 70% of every reported free-text item was in its training data; measured on selection gain the inflation is one-directional and about 31%. A disjoint-split retrain is in progress and will settle whether the free-text accuracy claim holds. Because the free-text arm holds 37.5% of the equal weight and is the load-bearing cell of every headline delta, this is the single largest open risk to the accuracy result. (v) The verifier is in-domain for the open-text families it selects on; cross-family transfer is positive but weaker (25% of the oracle gap on a from-scratch MedVLTinker verifier vs. 49% for Lingshu), and the “beats-the-32B” framing is seed-dependent. (vi) Best-of- N is dominated on FLOPs by escalation when—as here—the strong leg is cheap on both axes; its accuracy win on free text is real, but the deployable efficiency win is the router, and this verdict would change for an expensive/slow strong

model. (vii) The optimal-stopping draws are sequential, so on free text the controller wins FLOPs and energy but is slower at batch 1 than a batched fixed- N ; we report both concurrent and sequential latency and have not measured the overlap. (viii) The eight cells are not eight independent datasets—three benchmarks contribute a closed and an open cell from the same images—so equal weighting gives the free-text format 37.5% of the weight; the five-benchmark macro is reported as the counter-convention. (ix) MMMU-Medical is excluded on contamination grounds (the 7B scores 0.80 there against its own published 54.0 and beats its own 32B); under equal weight that exclusion is consequential rather than cosmetic, since MMMU would carry 1/9 of the average, and including it would raise the accuracy-max-vs-direct delta from +0.0128 to +0.0299. (x) All findings are on one model family for the headline; finding 1 generalizes across five, findings 2–3 in direction across two. The method always answers; no result relies on abstention.

Deployment implications. The practical reading is that a hospital deploying a medical VLM need not choose between a fast-but-weak 7B and an accurate-but-slow reasoning 32B: a format-aware router exceeds the reasoning model’s accuracy at roughly a tenth of its latency and energy, with a single knob to trade compute for a small accuracy gain and a per-benchmark guardrail on covered slices. The compute saving, though, is workload-dependent and should be sized on the deployment’s own traffic mix rather than on a suite average—on a stream dominated by easy multiple-choice questions it is close to the $0.49\times$ we measure pooled, and on a stream that mixes task types evenly it disappears. The components are individually simple—a margin threshold, a batched verifier score, a certified veto—and add no new model to serve beyond the two that a cascade already hosts. The negative results are equally actionable: effort spent engineering a cleverer escalation signal is misdirected, because the recoverability ceiling is a property of the model pair; the returns are in the structure (format-aware actions, an intermediate no-think tier) and in a small amount of verifier training. A natural next step is a slow think tier that pays off on a genuinely reasoning-heavy stream, and a verifier trained across more families to widen the free-text gains.

VII. Conclusion

In medical VQA, the accuracy–cost tension is largely a consequence of allocating test-time compute uniformly. Reasoning should fire only where it helps, sampling only where the format rewards it, and escalation only where the large model can actually recover the answer. A format-aware adaptive cascade that follows these rules lets a 7B model exceed the accuracy of every fixed way of using a 32B reasoning model—by +0.072 over always-reasoning and +0.013 over a single strong forward pass, under an average that weights every benchmark cell equally—while cutting roughly 90% of the reasoning baseline’s latency

and energy. We are equally explicit about what does not hold: the compute saving is concentrated on low-escalation multiple-choice traffic rather than uniform, so it is a real deployment saving on this traffic mix ($0.49\times$) and not a property that survives equal weighting ($1.20\times$); and on reasoning-heavy benchmarks the accuracy gain we can defend is an answer-format effect, not a reasoning-instruction effect. The negative half of the story is as useful as the positive: the escalation and selection signals are fundamentally bounded, so the leverage is structural and in a little training, not in a cleverer gate—and a suite average that lets one benchmark carry 79% of the weight will manufacture an efficiency claim that equal weighting removes.

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